

## **WEEKLY REPORT :**

**Project Title:** Self Driving RC Car

**Date:** 9/15/2026

**Instructor:** Dr. Nguyen

**TA:** Ruben Ramirez

### **Project Members:**

- Ethan Wells (Group Leader)
- Alexis Perez
- Ethan Bishop
- Abigail Duran

### **Weekly Summary (9/15/2026):**

- This week the team made progress on all fronts. Ethan Wells trained the traffic-sign model — a convolutional neural network on the GTSRB German dataset reaching about 95% validation accuracy — committed the model and training code, and ordered the CR2032 RTC battery and the KIOXIA NVMe boot SSD with a USB enclosure. Alexis Perez bench-tested the ultrasonic sensor's distance readings and accuracy and researched drive-motor options. Ethan Bishop tested the IMU for measurement accuracy, researched motor-controller options, and started drying filament to prepare the home 3D printer for chassis-component testing. Abigail Duran completed and refined the first-semester schematic around the Raspberry Pi Compute Module 5, resolved ERC issues to ready it for PCB development, and began hardware prototyping for the stationary test setup.

### **Proposed Plan for Next Week:**

- Ethan Wells: Prepare the LISA dataset (US traffic signs) for training; test the trained German-sign (GTSRB) model.
- Alexis Perez: Select and order motors; write code to integrate the ultrasonic sensor on the Raspberry Pi.

- Ethan Bishop: Select and order a motor controller; write code to integrate the IMU on the Raspberry Pi.
- Abigail Duran: Begin the PCB design and layout from the completed schematic; place major components and determine board dimensions; begin planning power and signal routing.

### **Weekly Contributions:**

- Ethan Wells:
- 9/10/2026 – Built and ran the first CNN training on the GTSRB German traffic-sign dataset in TensorFlow (3 hours)
- 9/14/2026 – Tuned the model to ~95% validation accuracy and committed the trained model (gtsrb\_model.keras) and training script to the repo (4 hours)
- Alexis Perez:
- Tested the ultrasonic sensor's distance readings and measurement accuracy
- Researched drive-motor options for the vehicle
- Ethan Bishop:
- Tested the IMU for measurement accuracy
- Researched motor-controller options
- Started drying filament to prep the home 3D printer for chassis-component testing
- Abigail Duran:
- Completed the first-semester schematic for stationary testing of the autonomous car system
- Built and refined the schematic around the Raspberry Pi Compute Module 5
- Resolved schematic/ERC issues and prepared the design for PCB development
- Began hardware prototyping for the stationary test setup
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### **Hour Tracker:**

| Name          | Hours For the Week | Cumulative Hours |
|---------------|--------------------|------------------|
| Ethan Wells   | 7                  | 12               |
| Alexis Perez  | 6                  | 9                |
| Ethan Bishop  | 6                  | 9                |
| Abigail Duran | 6                  | 11               |